Extraction of Region Contour in SEM Fractographs

***Abstract*—In the domain of material science, quantitative fractography is an analytical tool to study the characteristics of a fracture surface. The inception of Scanning Electron Microscope (SEM) has motivated the researchers toward the quantitative analysis of such surface. Due to fracture, new surfaces are evolved and voids are also formed. Extraction of such regions (surface/void) from SEM fractographs is of immense importance as it enables the subsequent characterization of the surfaces and the study of void distribution. To carry out the analysis, image processing tools are being applied by the researchers mostly on a case to case basis. Thus, well founded image processing technique to cater the specific need is still lacking. In this work, we have proposed a scheme to determine the closed contour of the regions denoting the surface or void. The proposed methodology relies on the systematic combination of basic techniques of image processing to accomplish the task in an automated manner. Experimental result indicates that the performance of the proposed scheme is satisfactory.**

Keywords: fractograph analysis, surface/void boundary extraction

I. Introduction

Quantitative fractography is an analytical tool which provides true estimates of the feature characteristics and topography of fracture surfaces. For about four decades quantitative fractography is being used as a tool in materials research. Early meaningful applications on steel [1] showed that new surfaces originating from fracture at varying conditions differ significantly in their quantitative geometric characteristics while often being indistinguishable in qualitative inspection.

Fracture of load bearing components while in service occurs for various reasons. The new surfaces evolved due to fracture carry the impression about the mode of fracture, which, in turn, is the manifestation of micro-structure of the materials and the loading conditions. Since the inception of Scanning Electron Microscope (SEM) researchers in the area of fracture of metals have attempted quantitative measurements on SEM fractographs. The SEM fractographs depict the topography, appearance and specific feature characteristic of fracture surfaces. The motivation to quantify fracture surfaces is to understand the precise role the micro-structure plays in the fracture process.

The common way to characterize a fracture surface is by quantifying the percentage area of the fracture surfaces covered by ductile and brittle fracture characteristics. The steps involved in the ductile fracture are void nucleation, void growth and void coalescence. The halves of these voids are represented by the dimple distribution on fracture surfaces. Increased ductility of a material is a consequence of delayed void nucleation and/or void growth. Therefore, it is important to understand the role of underlying metallurgical and mechanical (stress) variables that control the void nucleation and their growth.

Voids initiate and grow at inclusions, precipitates, and other second-phase particulate matter under the influence of plastic strains and hydrostatic stress [2], [3]. The role of inclusions and particles in initiating ductile fracture has been demonstrated for a variety of materials: oxidized copper alloys, maraging steels [4] quenched and tempered high-strength steels, low-strength steels [5], aluminum alloys [6] etc. The distribution, size, shape, type, and coherency of these constituents of the micro-structure play an important role in controlling void formation and eventual fracture [7], [8], [9]. For ductile materials, the engineering properties are determined by the interaction of stress and strain fields with the micro-structure of a material. The contribution of deformation processes in the development of voids has been well established. While the growth mechanisms can vary, say, with the temperature of testing [10] their nucleation invariably occurs where inhomogeneous deformation takes place.

Various attempts have so far been done to correlate the fracture characteristics with micro-structure. Summarizing the outcome of all such attempts it is revealed that the size, shape and distribution of the voids on fracture surfaces bears some correspondence with the size and distribution of second phase particles from where the voids are nucleated. Besides, void size, shape and distribution are also reported to depend upon loading rate and temperature [11], [12], [13]. In recent studies Das et el. [14], [15] showed good correlation between void morphologies with tensile properties of copper strengthened HSLA steel and 304 LN austenitic stainless steel. Results of all these studies thus indicate that very careful and systematic analysis of fractographic features can yield valuable information for correlations of micro-structural parameters with the micro-mechanisms of the fracture processes.

Past study reveals the significance of fractographic image analysis in the domain of material science. It requires extraction of the regions of different type, subsequent study of their characteristics and finally the quantification of the properties. Thus, the determination of region (surface/void) contour serves as the fundamental step and in the present work, we take up this task. Image processing techniques are also being deployed by the researchers [15] but the efforts are limited to the manual application of the general image processing tools on case to case basis. Moreover, the outcome depends on the acumen of the individual. It has motivated us to develop an automated process to serve the purpose. The paper is organized as follows. After this brief introduction, proposed methodology is presented in section II. Experimental results are shown in section III. Finally, the concluding remarks are put into section IV.

II. Proposed Methodology

It has been discussed in section I that fracture gives rise to different surfaces and deformation may lead to formation of voids also. Thus, the fractograph images show various surfaces and the voids present. Extraction of the region boundary is an important task that enables us to analyze the characteristics of the bounded region like whether it is a smooth surface or a void. It can further lead to quantitative study of void distribution. It has motivated us to develop an image processing methodology for extracting the closed contour of the surface/void.

Boundaries in a fractograph image is formed in number of ways. The most common one is that when two different surfaces meet each other. The other cases may be attributed to a significant void meeting the surface and in such cases, the surface boundary may also be taken as void boundary. Ideally, a void of considerable depth reflects a gradient in its intensity values with minimum intensity at the bottom and reaches maximum when it meets the surface. Small voids with negligible depth present in the surfaces do not reflect such gradual intensity variation. Identification of such small voids are not so difficult and in this work, such voids are taken as part of the enclosing surface. In general, at the boundary a sharp transition in terms of intensity values is expected. It has been observed in the fractograph images that the boundary pixels are of higher intensity values with white appearance and in other areas it is grayish/black in appearance.

The complexity in the surface undulation, diversity in their absorption and reflective property, wide variation in void depth and pattern lead to considerable deviation between the ideal scenario and the actual one reflected in the fractograph images. Thus, the determination of the closed contours of the regions can not be achieved in a straightforward manner and it becomes a critical problem demanding attention. The pixels of a surface/void around the boundary may also possess intensity values closer to those in boundary. Even the pixels in the surface/void may have considerable intensity variation. As a result a simple gradient based threshold or conventional edge detection techniques can not directly provide the desired output. Plenty of additional edges may come up. Moreover, the boundary will not be a closed one as all the pixels along the boundary may not have the desired high intensity values. Thus, the task requires the formulation of a set of steps that is less affected by the intricacies and can provide the optimal output.

Proposed methodology consists of a series of processing to achieve the goal of determining the closed contour of the surface. The steps are as follows.

- Binarization of the image.
- Formation of initial contour.
  - Thinning of the binary image.
  - Edge linking.
- Refinement of the contour.

A. Binarization of the Image

The gray-scale fractograph image is first binarized following intensity based thresholding. It has been discussed that in general, boundary pixels which form a fraction of the image are of higher intensity and the larger fraction arises out of the surface/void pixels are mostly black in appearance. Surface/void pixels may spread over a sizable range of intensity values but limited to comparatively lower side of the intensity values. Based on the intensity histogram, a threshold, th is chosen so that $B(i,j) = 1$ if $F(i,j) > th$ and $B(i,j) = 0$ otherwise. $F(i,j)$ and $B(i,j)$ denote the pixel at (i,j) location of the gray scale frcatograph image and the corresponding image after binarization.

Selection of th is crucial and it is chosen by studying the nature of the histogram carefully. Pixels, part of small voids and the bottom most level of the voids with considerable depth are of minimum intensity. Intensity of the void pixels as we move from bottom to the higher levels increases gradually. Intensity of the surface pixels also varies in intermediate range of the intensity levels. Intensity histogram may have two peaks in the extreme black and white regions signifying the presence of considerable void regions and large number of region surface regions (hence, the boundaries) respectively. Whether such peaks are present or not, the histograms show that in the intermediate range it is more or less bell shaped. A fractograph image and the corresponding histogram have been shown in Fig. 1(a) and (b) respectively.

In reality, the boundary pixels in a fractograph image are not of uniform intensity and show considerable variation. Thus, the selection of the local peak at higher intensity side as the threshold value may lead to highly porous boundary and thereby making it further difficult to form a closed contour. Moreover, such peak always may not be available. On the other hand, a low value for th will give rise to plenty of candidate pixels around the actual boundary, particularly for the voids with high depth. As a result, lots of false contour may be formed and a surface will be splitted into number of smaller regions.

To select the threshold, we concentrate on the bell shaped region of the histogram. To avoid the problem caused due to jaggedness in the histogram, intensity scale is divided into number of bins of small width. Corresponding to each bin centre, the average frequency in the bin is considered to form the smoothened histogram. Ignoring the peaks (if any) in two extreme intensity ranges, we restrict ourselves with in the intermediate range. It is assumed to follow a normal distribution with mean μ and standard deviation σ and th is taken as $\mu + 2\sigma$. It ensures that the majority of non-boundary pixels are discarded and most of the boundary pixels are retained. The image obtained after binarization is shown in Fig. 1(c).

B. Formation of Initial Contour

As it is evident in Fig. 1(c), the boundary obtained may have discontinuities and spurious pixels. At this stage, the target is to form a closed contour following the edge approximation

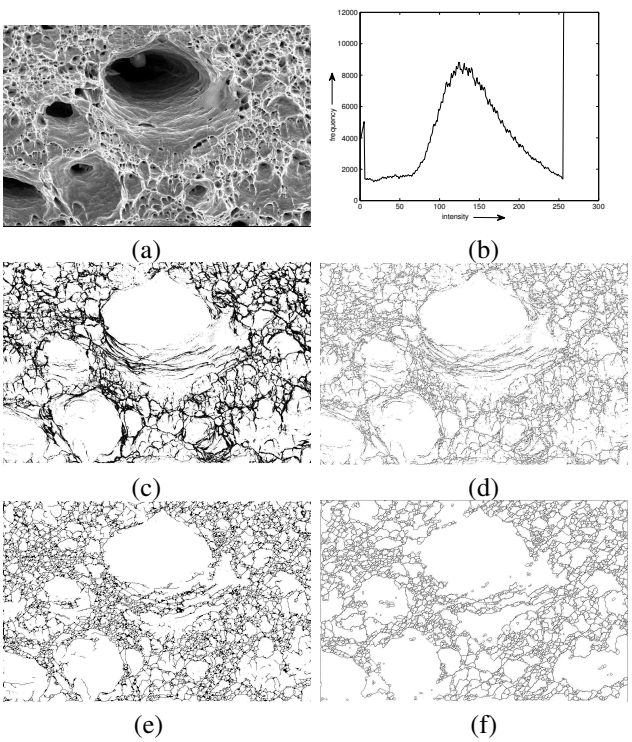


Fig. 1. Stepwise output of contour extraction process: (a) Fractograph image, (b) Intensity Histogram (intensity and frequency along x and y axis respectively), (c) Image after binarization, (d) Image after thinning (e) Initial Contour and (f) Final Contour

technique. As the detected boundary segments may be thick enough, we apply thinning as a preprocessing step to make the segments of single pixel width and thereby the complexity of the subsequent step is also reduced. Fig. 1(d) shows the image obtained after thinning.

In order to obtain the closed contour, we apply edge linking on the thinned image $B(i,j)$. Two nearby boundary pixels $B(i,j)$ and $B(i',j')$ with similar edge properties are joined together to form a long continuous contour. Corresponding to each boundary pixel, $B(i,j)$ gradient direction is computed by applying Sobel operator on the fractograph image $F(i,j)$ and it is taken as the edge property. In order to find out the similar pixel corresponding to $B(i,j)$, search is carried out among the unlinked boundary pixels with in its 8-neighborhood. The pixel with closest gradient direction is taken for linking. It may be noted that in case there is no unlinked boundary pixel present in the said neighborhood a break in the contour will occur. This may be attributed to considerable discontinuity present in the boundary obtained after thresholding. To minimize the occurrence of such breaks, the search for the point to be linked is carried out in a larger neighborhood of size $k \times k$ and the search area can be increased gradually to a maximum limit. But, a large search area may lead to the formation of closed contour considering the surface pixels sharing the boundary pixel like intensity value. In our experiment k is taken as 5 and gradually increased up to 7.

The linking procedure goes on iteratively till no further growth in the contour segments is possible. In each iteration, the thinned image is scanned in a particular direction and the boundary pixel encountered first is taken as the start point to initiate the edge approximation process. Once a segment can not grow further, the process tries to continue by scanning the image in the same direction to select another start point from the remaining boundary pixels. It goes on till no other start point is found by traversing in the same direction. The whole process is repeated for four different scanning directions (horizontally, from left to right or vice versa and vertically from top to bottom or vice versa) in each iteration. This ensures that a pixel, part of multiple boundary segment, gets the opportunity to establish the connectivity. It may also be noted that the growth of the segments at the end of an iteration may enable further linking in the subsequent iteration. In case, the boundary segments gain width, once again thinning process is carried out to obtain the output as shown in the Fig. 1(e).

C. Refinement of the Contour

After linking, surface/void contours obtained are mostly closed unless the boundary obtained after thresholding had a wide discontinuity. We may still have curve segments in the surfaces caused by the presence of false boundary pixels in the surface. Such curves in most of the cases are unlikely to form a closed contour. At this stage our aim is to prune out such open curves and to retain only the closed contours of the voids/surfaces.

we select a surface/void pixel in a detected region as a seed and region growing algorithm is followed. It recursively marks the pixels in the regions till the boundary pixels are encountered. The outer contour of the marked region provides the desired closed contour of the enclosed surface/void and also excludes the open and false boundary curves present in the region. The process continues for other regions by selecting another unmarked surface/void pixels as seed, if available. Thus, the final contour is obtained as shown in Fig. 1(f).

III. Experimental Results

A Titanium-stabilized Interstitial Free (IF) steel in the form of cold rolled sheets of 2.3 mm thickness received from TATA Steel Ltd., Jamshedpur, India, has been used in the present investigation. Two types of tests, tensile test and ratcheting test were performed on specimens made from the as received steel sheets. Strain controlled tensile test at a nominal strain rate of $10^{-3}\ sec^{-1}$ was done at room temperature (28^0C) under normal laboratory atmosphere in a computer controlled close-loop servo-hydraulic universal testing machine of $\pm 100kN$ load capacity, Instron 8501. Ratcheting tests were done by asymmetric cyclic stressing with different combinations of mean stress and stress amplitude at a stress rate of $250MPa\ sec^{-1}$ for different constant maximum stresses.

Fracture surfaces of specimens with and without prior ratcheting have been examined under scanning electron microscope (SEM) to find the effect of prior ratcheting on subsequent tensile fracture process. The specimens used for examining the features on fracture surfaces were obtained by sectioning the

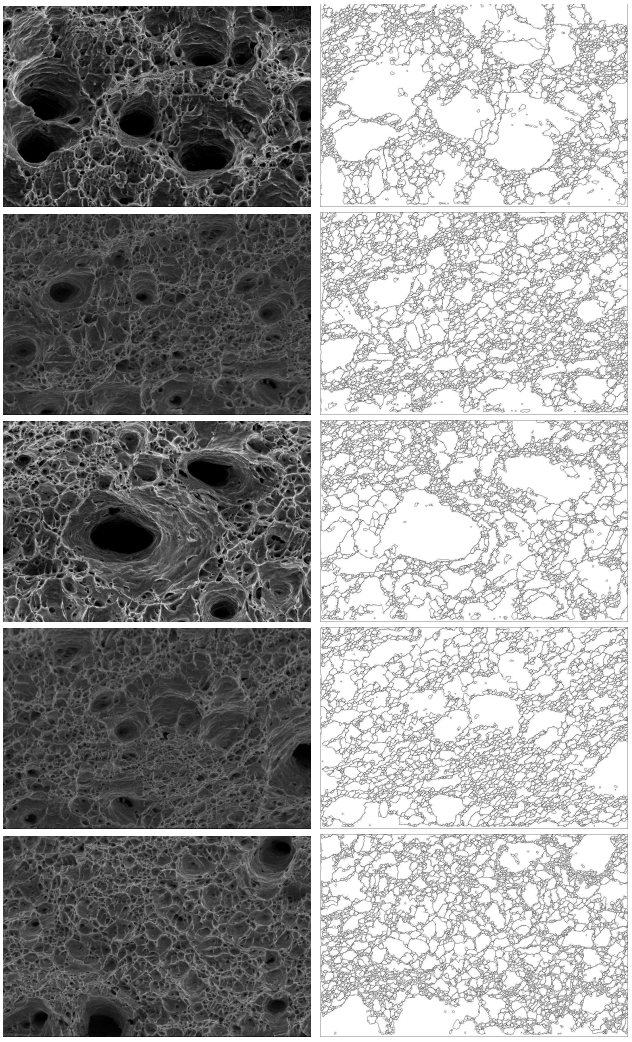


Fig. 2. Sample results: Fractograph image and extracted region contour

tensile fractured specimens (with and without prior ratcheting) on planes perpendicular to the tensile axis leaving a distance of $10mm$ from the fracture surfaces. Before inserting the sectioned specimens in the vacuum chamber of SEM the fracture surfaces were ultrasonically cleaned and observed in a Jeol make, Japan, Scanning Electron Microscope (JSM6360) under secondary electron imaging mode with an accelerating voltage of $20kV$ and under condition of zero tilt. The brightness and contrast of the images were controlled by the in-built image controlling software of the microscope. However, manual intervention was also done in controlling the brightness and contrast, wherever needed. Secondary electron images of the fracture surfaces from each of the sectioned specimens taken at three different magnifications ($\times 500$, $\times 1000$ and $\times 2000$) were digitally recorded in a computer interfaced with the microscope. In total 68 secondary electron images were taken. The fractograph images thus obtained have been used in this work.

Representative results have been shown in Fig. 2. It has been observed that proposed methodology was successful in extracting the contour of desired regions(surface/void) that will enable us to carry out further analysis of region characteristics and quantification of the feature in future.

IV. Conclusion

Due to fracture, new surfaces emerge and voids are formed in the material. Characterization of such surfaces and the study of void distribution is very important in quantitative analysis of fractograph images. Extraction of the regions (surface/void) and their contour serves the first step to carry out the subsequent analysis. Mostly, the researchers accomplish the task using a general image processing tool on a case to case basis. In this work, we have provided an image processing technique based comprehensive scheme for automatically extracting the closed contour of the regions. Experimental result indicates that the scheme works well for a wide range of images.

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